Fiscal Year:	FY 2019	**Task Last Updated:**	FY 08/23/2019
PI Name:	Lee, Michael Ph.D.		
Project Title:	Microbial Evolution and Transmission Aboard the ISS: Inferring Mutation Rates, Assessing Pangenomes, and Tracking Microbiome Transmission between Astronauts and the Space-based Built Environment		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-MOBE. Appendix B: Post-Doctoral Fellowships to Study the Microbiome of the ISS as a Built Environment
Start Date:	09/13/2018	**End Date:**	07/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Everroad, Craig	**Contact Phone:**	650-604-4997
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 6/15/20)		
Key Personnel Changes/Previous PI:	August 2019 report: There have been two personnel changes on the project as a whole. Dr. Alexander Voorhies was initially a co-investigator as he was heavily involved with the Astronaut Microbiome Project under Principal Investigator (PI) Dr. Hernan Lorenzi at the J. Craig Venter Institute (JCVI). Alexander has since moved on from JCVI and as such is no longer involved with this project. This hasn't hindered the work we are doing in integrating the astronaut microbiome data with our ISS isolate data. Another change is that I have closely partnered with Dr. Aubrie O'Rourke. Aubrie was awarded funding from the same MoBE call (hers through Sloan). She works out of JCVI in San Diego, CA where co-I Christopher Dupont is located, who connected us after we were both funded. Aubrie is very much "wet-lab" biologist, who was focusing on doing physiology and phenotypic assays working with Burkholderia isolated from the ISS, while I am very much a bioinformatician who was focusing on doing computational analyses with Staphylococcus and Ralstonia isolated from the ISS. We have since merged our isolate libraries, both working on all, and all of the science coming out of it is for the better. Neither of these personnel changes involves any budgetary changes.		

COI Name (Institution):	Everroad, Craig Ph.D. (MENTOR: Bay Area Environmental Research Institute/NASA Ames Research Center) Dupont, Christopher Ph.D. (J Craig Venter Institute, Inc.)
Grant/Contract No.:	80NSSC18M0110 (note funded under BAERI grant)
Performance Goal No.:	
Performance Goal Text:	
Task Description:	POSTDOCTORAL FELLOWSHIP NOTE: This project is funded by NASA as a result of a Space Act Agreement between NASA and the Sloan Foundation, for Collaborative Research Proposal for Microbiology of the Built Environment (MoBE). The purpose of this Space Act Agreement is to enhance the ability of NASA and the Sloan Foundation to support scientific research and the dissemination of information about the effect of spaceflight on microbial ecosystems in spacecraft such as the International Space Station (ISS) through sponsorship of postdoctoral fellowships. NASA is providing the isolates for all studies. OVERVIEW The proposed work will offer genetics-based avenues to gain insight into all 3 of the guiding questions as listed in this solicitation, but will directly address: “Does long-duration spaceflight alter normal rates of evolutionary change?” We propose to perform whole-genome sequencing of 50 isolates from the International Space Station (ISS) focused within the genera Staphylococcus and Ralstonia. These ISS-isolate genomes will be analyzed within the context of both an exhaustive, Earth-based reference genomic library constructed for each genus, and ISS astronaut microbiome metagenomic data. Co-Investigator Alexander Voorhies and collaborator Hernan Lorenzi are currently working with ‘before,’ ‘during,’ and ‘after’ microbiome metagenome from 9 ISS astronauts that were onboard the space station for various lengths of time spanning 2013–2016. Our unique position of being able to integrate microbiome sequencing data from another NASA effort with newly sequenced genomes of ISS isolates offers an unprecedented opportunity to explore many Space Biology evolutionary questions at the molecular level. OBJECTIVES/METHODS Genera Staphylococcus and Ralstonia were selected based on several criteria including ubiquity within human microbiota, potential pathogenicity, and their abundant representation within Earth-based reference databases, the ISS archive of available isolates, and our currently in-hand astronaut microbiome metagenomic data. We will investigate many metrics of molecular evolution aboard the ISS by directly addressing the following questions: 1) Are any particular genetic elements (i.e., genes/gene cassettes/operons) disproportionately represented in ISS isolates as compared to their Earth-based, nearest phylogenetic relatives? 2) Are there different propensities for specific individual genome mutations (i.e., single-point mutations such as substitutions/insertions/deletions, non-synonymous to synonymous substitution ratios, transition to transversion ratios, and genetic rearrangements) for organisms in the space-based built environment? 3) Which specific genes (and codons within these genes) are under positive, negative, or neutral selection aboard the ISS, but not within terrestrial references? 4) Can we track microbial transmission from an individual astronaut to the built environment, and/or the ISS serving as a vector and contributing to the microbiome of resident astronauts? 5) As assessed via comparison to metagenome-assembled genomes reconstructed from ISS-astronaut microbiomes, how representative are the isolate genomes recovered from the ISS of the actual microbiota of ISS astronauts?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	SIGNIFICANCE Microorganisms are not only integral to human health, but are also inexorable and necessary components of any self-contained elemental cycling, food production, or water system, and longer-duration human space exploration will require all of these systems. To design safe, built-environment microbial ecosystems to support these life-support systems requires not only understanding short-term, “plastic” microbial responses to the space environment, but also longer-term propensities for microbial evolution and adaptation. This work will leverage newly developed genetics-based computational tools, technologies, and approaches to generate the first comprehensive, whole-genomic database of microorganisms isolated from the space-based built environment while addressing fundamental questions about evolution aboard the ISS.
	This work employs a multitude of physiological and bioinformatic approaches to investigate microbial isolates recovered from the International Space Station (ISS). We do not currently have the capabilities to design and implement a decades-long evolutionary experiment aboard the ISS with proper controls, but the archive of microbes that have been isolated from the ISS over the course of 2006–2015 provides an invaluable collection that can be explored after the fact. We are utilizing this isolate collection to ask many questions about these microbes in contrast to their nearest relatives that have not been aboard the ISS (e.g., gene-enrichment/depletion, antibiotic resistance profiles and genetic complements, plasmid composition, rates of single-nucleotide and single-amino acid variability as a function of core phylogenetic distance). We additionally are combining this dataset with another invaluable dataset, metagenomic data from the Astronaut Microbiome Project, to try to track if there is any evidence of microbial transmission from the ISS microbiome to the microbiome of resident astronauts. We have successfully received, cultured, sequenced, and assembled the genomes of 114 isolates recovered from the ISS – 24 of genus Burkholderia, 37 Ralstonia, and 53 Staphylococcus. All 3 of these genera are common human microbiome members and potential pathogens. Work thus far has given us some idea of the variability of these organisms and their plasmid content, how they relate to their nearest Earth-bound relatives phylogenetically, gene-content-wise, and in terms of their susceptibility to antibiotics. For an example in terms of variability, we recovered 24 isolates from 2

Task Progress:	species of Burkholderia, B. cepacia and B. contaminans, which were isolated over a 5-year period. Within species, these all form monophyletic clades based on phylogenomic trees with all available references, and all demonstrate greater than 99% average nucleotide identity based on successful genome alignments of greater than 95%. This reveals all members of each of these 2 species are of a highly similar genomic lineage and may have all stemmed from the same or very similar founding populations, rather than being a mixed assemblage. A pangenomic analysis (considering the entire genetic complement of some defined group of organisms) incorporating just the 19 ISS-derived B. cepacia, for instance, demonstrated that they all shared greater than 90% of the same genes. And the identification of putative plasmids revealed that this small proportion of genetic variability that exists appears to be harbored on a highly variable population of plasmids, carried alongside a highly conserved and stable chromosomal population. When incorporating these 19 ISS-derived B. cepacia into a pangenomic analysis with 136 available B. cepacia reference genomes allowed us to identify 265 functions significantly enriched in our ISS isolates as compared to the references (several annotated in 19/19 of our isolates and 0/136 of the references), and 98 functions significantly depleted in our isolates. It is important to keep in mind however that because these differences co-vary with genome phylogeny, it is not yet knowable if these particular enriched traits are a consequence of time spent on the ISS, or just a result of a more general evolutionary divergence through time. In an effort to find evidence of microbial transmission from the ISS microbiome to the microbiome of an astronaut, we have also integrated these newly sequenced ISS-isolate genomes with data from the Astronaut Microbiome Project. This includes astronaut microbiome metagenomic data from multiple timepoints before, during, and after a mission on the ISS. Through recruiting the short reads from the astronaut microbiome metagenomic data to our ISS-isolate genomes, we have identified at least genome so far where there was no recruitment at the four timepoints pre-flight from an astronaut's nasal microbiome, strong recruitment across the entire genome when sampled on days 90 and 180 onboard the ISS, and then near-zero recruitment after returning to Earth. We see two possible explanations for this. One is that something similar enough to recruit to this isolate colonized the astronaut's nasal microbiome when they were aboard the ISS, and then began being outcompeted when the astronaut returned to Earth. The second is that something similar enough to recruit to this isolate genome was already present in the astronaut's nasal microbiome, but was at too low of a relative abundance to be captured before their time aboard the ISS. Then the ISS environment was conducive to it becoming more abundant, which let us capture it with this approach, and after returning to Earth it may have returned back down to its lower abundance. We think it is the former, transmission, but we cannot at this point rule out the possibility that it is the latter. Either way something interesting is going on. We are in the process of writing up this work.
Bibliography Type:	Description: (Last Updated: 01/19/2022)
Articles in Peer-reviewed Journals	Lee MD. "GToTree: a user-friendly workflow for phylogenomics." Bioinformatics. 2019 Oct 15;35(20):4162-4. Published online 2019 Mar 13. https://doi.org/10.1093/bioinformatics/btz188 ; PMID: 30865266; PMCID: PMC6792077 , Oct-2019
Articles in Peer-reviewed Journals	Lee MD. "Applications and considerations of GToTree: a user-friendly workflow for phylogenomics." Evol Bioinform Online. 2019 Jul 25;15:1176934319862245. eCollection 2019. https://doi.org/10.1177/1176934319862245 ; PubMed PMID: 31384124; PubMed Central PMCID: PMC6659180 , Jul-2019